

RESEARCH

Open Access



Analyzing the trend of government support for cloud computing usage in e-government architecture

Muhammad Younus^{1,2*}, Eko Priyo Purnomo¹, Achmad Nurmandi¹, Dyah Mutiarin¹, Halimah Abdul Manaf³, Fathiya Mumtaz¹ and Tiara Khairunnisa¹

Abstract

The revolution in improving services to the community carried out by the current government is genuine. It is not easy for government organizations, especially local governments, to directly implement e-government services in full. One solution that is considered appropriate and can solve these problems is the application of cloud computing to support e-government services in local governments. The advantage of cloud computing for e-government is that it can increase security. In contrast, cloud-based storage can be valuable because data stored in cloud computing is guaranteed secure and various regulations and standards of information safety practices. We propose a systematic literature review approach to analyze trends in studies on cloud computing of e-government. This research is a type of descriptive qualitative research using a bibliometric approach. To assess the trend of cloud computing in e-government, we use CiteSpace's latest bibliometric software to achieve a comprehensive knowledge overview of cloud computing in e-government. The Findings of this paper reveal a dynamic scenario influenced by advancing technological progress and administrative priorities. Across the globe, governments are progressively acknowledging the capacity of cloud computing to improve the effectiveness, accessibility, and scalability of e-government services. Overall, challenges persist, spanning from concerns regarding data safekeeping and privacy, but it also signifies a strategic transition towards harnessing digital technologies to provide more agile, citizen-focused public services.

Keywords E-government, Cloud computing, E-architecture, Bibliometric analysis

Introduction

Information technology develops rapidly and extensively and requires more time to predict the time. The change was also unnoticed and has rapidly changed organizational behavior and is disruptive, so whether you like it or not, the organization is forced to carry out radical

reconstruction (significant improvements). It takes much work for organizations to avoid the era of innovation from the dynamics of information technology that continue to develop. The organizational perspective in carrying out his business process slowly began to shift, influenced by the development of information technology [1]. In the last few years, we have witnessed and felt that the government has been trying to improve the service to society. One form of effort for the government to improve services to the community is through the implementation of e-government services [2].

The implementation of e-government is felt to have many benefits for citizens and the government. For society, e-government is used to manage data, improve public services, and expand communication channels

*Correspondence:

Muhammad Younus

mohammedyounusghazni@gmail.com; m.younus.psc23@mail.umy.ac.id

¹ Department of Government Affairs and Administration, Universitas

Muhammadiyah Yogyakarta, Yogyakarta, Indonesia

² Department of Product Research and Software Development, TPL

Logistics Pvt Ltd, Karachi, Pakistan

³ UUM College of Law, Government, and International Studies, Universiti Utara Malaysia, Kedah, Malaysia

[3]. In contrast, for the government, the implementation of e-government can improve service productivity, model development services, increase the knowledge of government officials, and increase the effectiveness and efficiency of services to the community. However, there are many benefits to be gained from implementing e-government, but not most minor perceived obstacles [4]. E-Government, as a concept implementation of information technology in the government sector, will increase the efficiency and effectiveness of the government administration.

Local government started utilizing Information Technology and Communication (ICT) as intensively appropriate with the Instruction of the President of the Republic of Indonesia Number 3 of 2003; however, it was sometimes constrained by the budget and procurement time. It takes time a relatively long time to procure temporary new ICT infrastructure [5]. Existing ICT has yet to be utilized maximally. There are many challenges faced by local governments in developing e-government services, including inadequate infrastructure, limited human resources, budget constraints, problems with software licenses and their support, difficulty migrating services, integration and management between software and hardware, as well as various things that often cause failure in the development of e-government programs [6].

Therefore, the right solution is needed to overcome these challenges. When viewed from the advantages, this will be a reasonable consideration for governments to use cloud computing, where one of the government's focuses is service-oriented public. The government can focus more on its "public service" and related affairs another government without the need to bother supporting the infrastructure behind it. Cloud computing can consolidate ICT infrastructure, simplify management, maximize the utilization of computing resources, and ultimately save on the operating costs of ICT [7]. According to [4], the definition of Cloud Computing is a model that allows for ubiquitous (anywhere and whenever), convenient, on-demand network access to computing resources (e.g., networks, servers, storage, applications, and so on). services) that can be quickly released or added.

Cloud computing is a cloud-based storage system accessed online so that users can directly access stored data with an internet connection. Judging from the added value in the cloud computing storage system, it will be advantageous for the government system if it uses it. Cloud computing is divided into three parts: the characteristics of cloud computing, cloud computing service models, and cloud computing

implementation models [8]. The characteristics of cloud computing consist of on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured services. Service models of cloud computing consist of software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS). The model's cloud computing implementation consists of public cloud, private cloud, hybrid cloud, and community cloud [9].

Along with the development of cloud computing technology, several government agencies are planning to use cloud computing, and even institutions have already started to take advantage of this technology [10]. Utilization of cloud computing technology can help the government integrate the data; besides that, from the side, users will be more cost- and time-efficient, risk-free, and also do not need to provide room or place again (space). E-government services in the village scope are activities directly related to the community in the form of village administrative service activities [11]. Village administrative service activities are activities carried out routinely by village officials for every community in need.

Form of service village administration, such as making various cover letters to sub-districts, districts, or even provinces, is an activity that should be uniform for every village within the scope of one sub-district and the sub-district, which stores all demographic data [12]. Also, Cloud technology implementation computing as a Digital Asset Management (DAM) is influenced by requirements, cost-effectiveness, security, and system reliability factors. Cloud utilization computing as a Digital Asset Management (DAM) at the Presidential Palace Yogyakarta is expected to better influence online collaboration between employees and facilitate coordination across distances by optimizing existing information technology [13].

Literature review

'E-Government' in Contemporary World

The extent to which various governmental bodies have implemented automation and digitization in their operations and offerings represents the fundamental concept of the Electronic Government. To what extent have these entities progressed in implementing e-government initiatives to execute crucial national development endeavors and advance their objectives successfully? The exploration of E-Government Solutions deployed across numerous nations not only serves as a yardstick but also as a blueprint for development that enables each country to glean insights from others and adopt the most efficient strategies for addressing challenges and surmounting obstacles.

E-Government Initiatives are anticipated to enhance civic engagement in addressing challenges from national governments. The proliferation of information technology and the ability to retrieve information online is poised to prompt citizens in the E-Government Development Index of the new Member States to undertake measures aimed at overseeing their governments through e-government tools, ultimately leading to a reduction in corruption. The evolution of e-government has brought domains and data previously under limited external scrutiny to light, granting the public a clearer insight into the administrative sphere once shrouded in secrecy. The capacity of e-government to enhance the monitoring and regulation of governmental activities is projected to mitigate corruption risks in the public sector.

Digital technologies promise to enhance good governance, determine whether nations leverage rapid technological progress to bolster prosperity and inclusivity and explore how advancements in electronic government and good governance can foster long-term growth [14]. The challenge lies in initiating effective reforms to optimize digital advantages while remaining resilient to potential disruptions. The digital landscape is swiftly evolving; therefore, neglecting to adapt complementary digital components, such as regulations, skills, and institutions, may lead to escalated opportunity costs. Failure to enact reforms will lag behind those who do not embrace change. Enhancing the synergy between technology and its complements is now more crucial than ever.

Understanding the concept of ‘cloud computing’

Previous research [15, 16] showed that several government agencies are looking forward to reaping the benefits of integrating cloud computing into their organizational infrastructure. Cloud computing is one of the best solutions to overcome the lack of IT infrastructure that prevents e-government from meeting the requirements for high quality [17]. In addition, implementing cloud computing will present a constructive opportunity to cut costs while developing the electronic government system [17].

Previous studies by Ji & Liang [18], have shown that modern information technology trends are moving toward cloud computing, which promotes sharing, virtualization, and distributed collaboration. Therefore, the factors that led to the initial adoption of cloud computing have significance in academic circles. Implementing cloud computing technology within the government will facilitate the development of the information sector and the capacity for independent innovation regarding cloud computing products [19]. Previous studies by Wang et al., [20] suggest multiple adoption theoretical models from the viewpoint of the system to investigate the factors that

influence the adoption decision for e-government cloud services.

Previous research by Alkhawaldi et al., [21] showed that developing and even developed countries must create e-government to reduce costs and grow sustainably in this crucial economic situation. The best way to accomplish this is through green and inexpensive technology like cloud computing. Without a shadow of a doubt, nations’ participation with one another on various technical and legal issues is the code key to achieving e-government based on cloud computing as quickly as possible [22]. In addition, it can eliminate or significantly reduce the difficulties and obstacles currently in the way. As a result, an electronic government has been established, and many people are interested in and actively participating in its services [3].

This has also been explored in prior studies by [23] Cloud computing has been of tremendous assistance to the nations of the Middle East, not just in the realm of governance but also in terms of the region’s overall economic growth. The rise in internet use and population in the Middle East has led to increased demand for and the application of cloud computing [12]. This has helped the region advance economically because of the region’s confidence in the dependability and safety of cloud storage and computing. The goal of cloud computing in Europe and the Middle East is to increase communication, data sharing, and data access. This is accomplished by sharing electronic information and data across organizations, government agencies, and departments [24].

Application of cloud computing in e-government

Prior research by Hwang et al., [25] it has been suggested that a professional e-government framework be created to reduce the money and effort organizations invest. The benefits of cloud computing inspired the invention of such a framework. In addition, utilizing a private government cloud has avoided the restrictions and limits associated with cloud computing [26]. According to its capabilities, E-government services can also be migrated, imitated, developed, and maintained with its help [13]. In addition to that, it is aware of the interoperability, integration features, replication, manageability, flexibility, and accessibility of integrated database centers.

Previous studies [17] have emphasized that in the not-too-distant future, many information systems will be supported by cloud computing due to the rapid evolution of technology. This is because its strengths in information safety and communications outweigh its weaknesses. Cloud computing, which introduces a whole new design, will eliminate all of the flaws in the existing information

systems. By taking into account the numerous benefits of cloud computing, such as its low cost, independence from a physical location, scalability, and safety of information systems, which are the primary pillars of success in E-government, we can create an E-government that caters to a wide variety of audiences within a database that is comprehensive, effective, and efficient [6]. In this current economic climate, emerging countries and industrialized nations urgently need to establish an electronic government to lower implementation costs, promote economic growth, and develop sustainably. The use of cloud computing technology, which is both environmentally friendly and cost-effective, is the most effective way to accomplish this goal [27].

A series of recent studies by [28]. Has indicated that computing in the cloud is actively working to help break down barriers that prevent many governments from entering a new phase of collaboration and partnership, sharing services, and pooling resources. It provides an efficient way for citizens to share information, which helps reduce the effort required to provide services while improving financial management and lowering costs [19]. In addition, the significance of embracing new technologies in implementing e-government, such as cloud computing, which supplies the overarching strategy and specific methods to manage e-government content in general, was emphasized [15, 16].

Countries' Participation with one another on technological and legal issues is essential for realizing e-government based on cloud computing. Moreover, it can overcome the difficulties and obstacles that lie ahead. The majority of prior research by [3] The implementation of e-government in Indonesia, a developing country, is still plagued by several challenges, and the state of affairs can be described as less than ideal. This is demonstrated by the government's website's place in the worldwide e-government ranking. In general, the electronic government in Indonesia could be more efficient and effective [29]. The approach of making substantial investments in each region is responsible for the inefficiency of the budget. In addition, human resources in government institutions are regarded as being limited, particularly in terms of the capabilities of the utilization of ICT [23]. The model now being used to deliver e-government services in Indonesia needs to be revised to improve it.

Research method

In analyzing the problem, this study is based on the qualitative method as the research type. According to the qualitative descriptive method is a research method based on post-positivism philosophy used to research natural object conditions (as opposed to experiments)

where the researcher is the key instrument. Data collection techniques are carried out in triangulation (combined), data analysis is inductive or qualitative, and the results of qualitative research emphasize meaning rather than generalization. Qualitative descriptive research aims to describe, explain, explain, and answer in more detail the problems to be studied by studying an individual, a group, or an event as much as possible.

Research approach

In qualitative research, humans are research instruments, and the writing results are in words or statements based on the actual situation. Google Scholar and Scopus are used as data sources in this study. Google Scholar is an academic research tool that is part of Google Search, the most popular and effective search engine on the internet. An Indian-born computer scientist named Anurag Acharya was the brains behind the creation of Google Scholar in 2005. In its most basic form, Google Scholar compiles Internet webpages that appear like articles or other scholarly documents. As a result, it has developed into a portal through which one can access scholarly content on the Internet. Daily, more academic knowledge is made available online, and users continue to discover new reasons why they require access to this information [30]. Elsevier's Scopus is a database containing full-text links, abstracts, and indexes, and Scopus produces it. The bird known as the Hamerkop (Scopus umbrella), said to have exceptional navigational abilities, inspired Scopus. The database was constructed for two years and involved the participation of 21 research institutes and more than 300 academics and librarians. The verbal and non-verbal feedback provided by these researchers and librarians was evaluated and incorporated into the development of the product [31] (Fig 1).

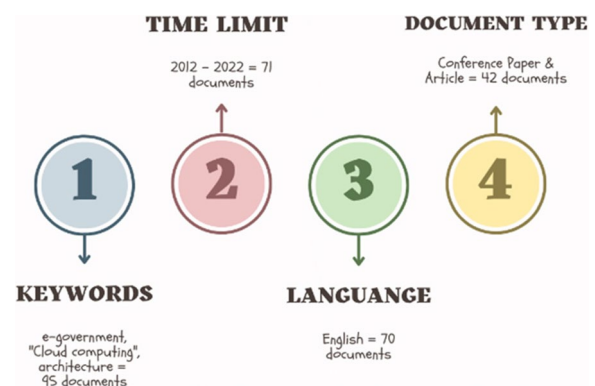


Fig. 1 Illustrates the representation of Data mining step on research publications within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication. Source: Created by the author

Data collection technique

For the data collection techniques, the writers do four steps to filter the keywords to get the data. The first step with results 95 documents is to put e-government in three different ways (e-government, "E-government," and "Electronic government"), then put the word "Cloud computing," last is the word architecture in two different ways (architecture and architectures) then it will look like this (TITLE-ABS-KEY (e-government) OR TITLE-ABS-KEY ("E-government") OR TITLE-ABS-KEY ("Electronic government") AND TITLE-ABS-KEY ("Cloud computing") AND TITLE-ABS-KEY (architecture) AND TITLE-ABS-KEY (architectures)).

The next step is to sort out the publishing year by ten years (2012 – 2022) so later, it will look like this: (TITLE-ABS-KEY (e-government) OR TITLE-ABS-KEY ("E-government") OR TITLE-ABS-KEY ("Electronic government") AND TITLE-ABS-KEY ("Cloud computing") AND TITLE-ABS-KEY (architecture) AND TITLE-ABS-KEY (architectures)) AND PUBYEAR>2011 AND PUBYEAR<2023 AND PUBYEAR>2011 AND PUBYEAR<2023. The third step is to sort out the language by just using English only so it will look like this (TITLE-ABS-KEY (e-government) OR TITLE-ABS-KEY ("E-government") OR TITLE-ABS-KEY ("Electronic government") AND TITLE-ABS-KEY ("Cloud computing") AND TITLE-ABS-KEY (architecture) AND TITLE-ABS-KEY (architectures)) AND PUBYEAR>2011 AND PUBYEAR<2023 AND (LIMIT-TO (LANGUAGE, "English")).

The last step is to filter the source type by only using Conference Paper and Article as the source of references so, in the end,, it will look like this (TITLE-ABS-KEY (e-government) OR TITLE-ABS-KEY ("E-government") OR TITLE-ABS-KEY ("Electronic government") AND TITLE-ABS-KEY ("Cloud computing") AND TITLE-ABS-KEY (architecture) AND TITLE-ABS-KEY (architectures)) AND PUBYEAR>2011 AND PUBYEAR<2023 AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (DOCTYPE, "cp") OR LIMIT-TO (DOCTYPE, "ar")) (Fig. 2).

Data analysis technique

Writers utilize CiteSpace software [32] for data analysis. The program identifies subject areas and citation hot-spots and divides the network into clusters. CiteSpace simplifies WoS data for easier understanding and interpretation. It marks clusters using citation terms, collaboration patterns, and cooperation domains. The software supports co-citation network analysis with a time-slicing methodology. Synthesis involves taking snapshots of the network at critical nodes. CiteSpace assists in

systematically reviewing and summarizing research. It aims to automate laborious tasks for users and focus on problem-solving. Network analysis and visualization are vital to CiteSpace [33]. It explores the intellectual landscape of a knowledge area. CiteSpace was created to address inquiries about a knowledge domain. It represents a collection of bibliographic records for relevant publications. Users must assemble a relevant and representative dataset to answer questions about a knowledge domain. CiteSpace simplifies the process of answering inquiries regarding knowledge domain organization and operation.

Results and discussion

Trend publication

Figure 3 illustrates the quantitative representation of research publications about using Cloud Computing within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period.

Figure 3 shows that for the past ten years, from 2012 until 2022, the publication trend regarding cloud computing e-government architecture sharply decreased. In 2012, the topic had 17 publications, and then for two years in a row, in 2013 and 2014, the publication went downhill, but the numbers are stable with nine publications. Next, in 2015, the number of publications decreased by 2 points, so there were only seven publications in that year. From 2016 until 2018, the publications went up and down by only two publications in 2016; then it went up by six publications in 2017, and then it went down again in 2018 with only one publication published in a year. Later, in 2019, the number of publications rose to 4; in 2020, it increased to 5. In 2021, the publications were only four a year, but a year later, in 2022, it increased to 6.

Authorship and highest citation

Figure 4 illustrates the Authorship representation of research publications based on the highest citations pertaining to the utilization of Cloud Computing within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period of time.

In Fig. 4, there are ten authors who have written the most regarding cloud computing e-government architecture. The person who wrote and published the most about the topic is Zwattendorfer, B., with three articles. Next is Biancofiore, G., Hashemi, S., Hashemi, S.Y., Leone, S., Liang, J., Monfaredi, K., and Tauber, A., who wrote and published two articles in total. Last is Dahab, A.H. and Ahmad, M., who wrote and published 1 article each.

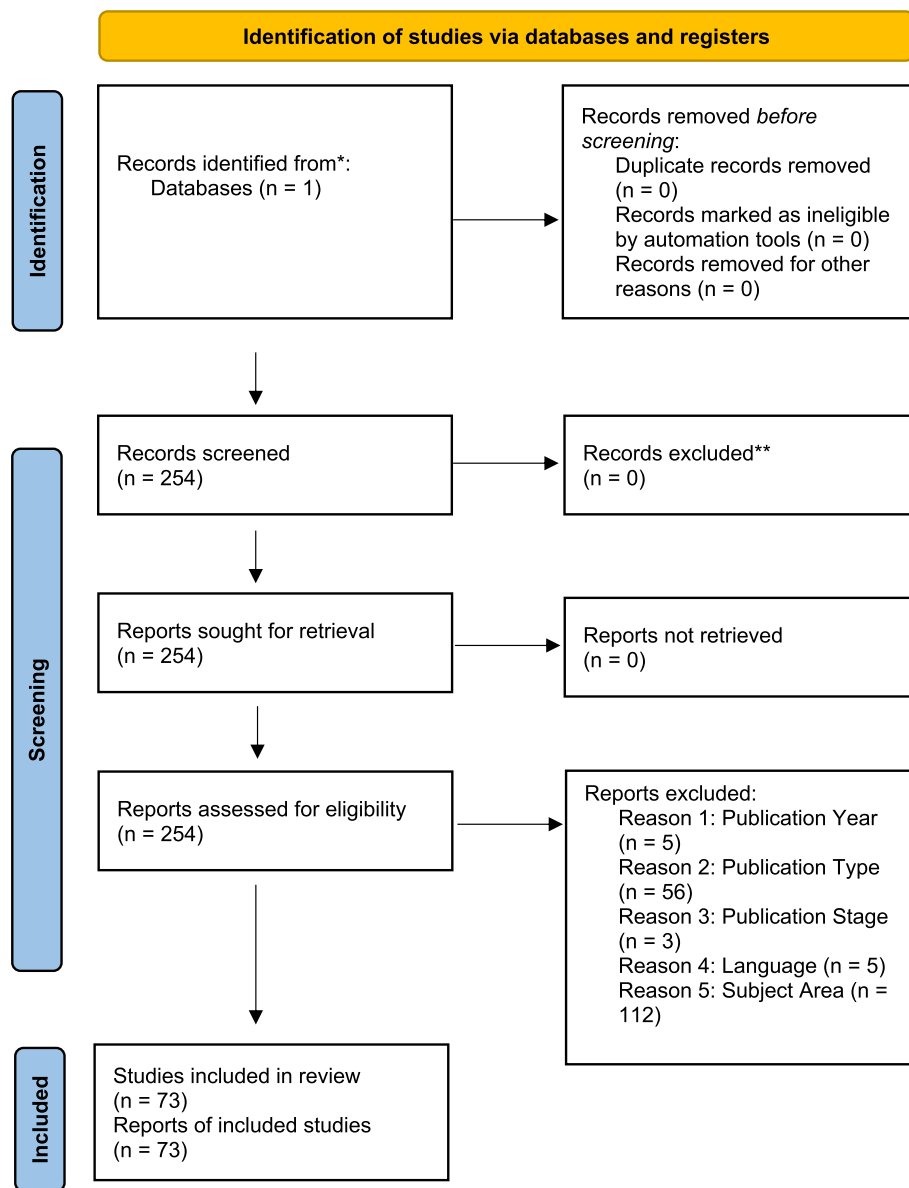


Fig. 2 Illustrates the representation of PRISMA diagram flow on research publications within the realm of E-Government. The subsequent flow diagram will delve into the patterns and fluctuations observed within the literature review. PRISMA diagram flow source: created by the author

Distribution and network citation by country

Figure 5 illustrates the representation of research publications based on the country pertaining to the utilization of Cloud Computing within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period of time.

Figure 5 shows that, with 10 out of 25 countries, China has the highest number of citations, with 14 publications. Followed by India with eight publications and then Italy with four publications. Next is Austria, which has three

publications, followed by Greece, Indonesia, Iran, Luxembourg, Pakistan, and South Korea, which have two publications.

Publication venues by subject area

Figure 6 illustrates the representation of research publications based on the Subject area pertaining to the utilization of Cloud Computing within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period of time.

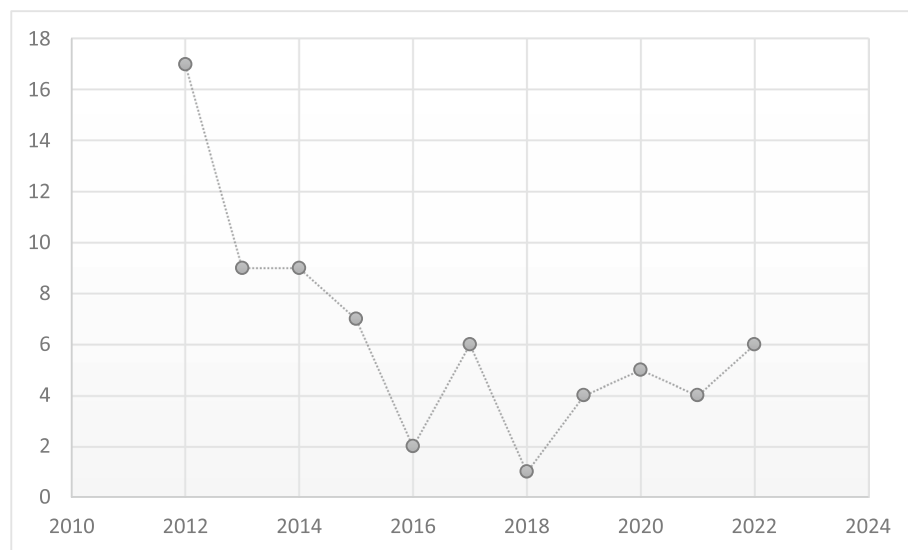


Fig. 3 Publication trend by years. Source: scopus database

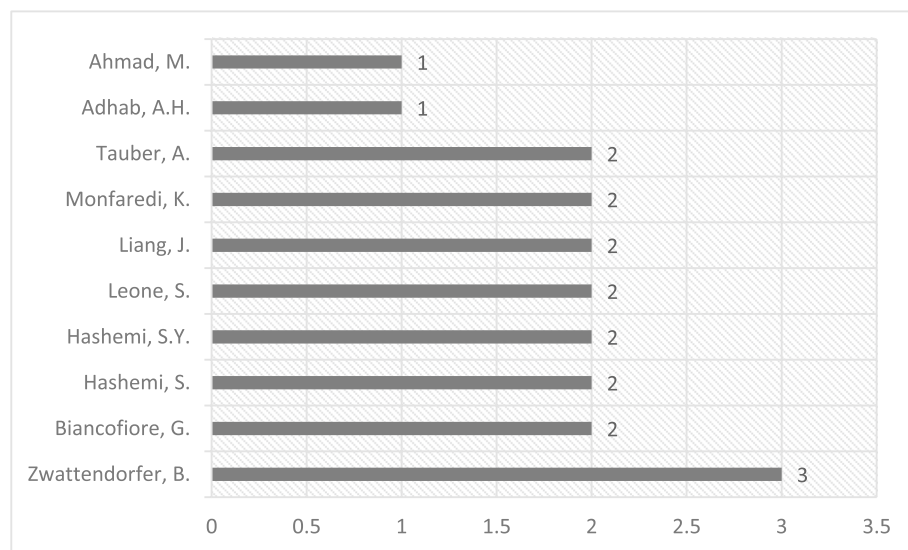


Fig. 4 Publication by authors. Source: scopus database

Figure 6 shows that, with the topic related to electronic government, cloud computing, and architecture, ten subjects with the highest citations are connected to it. Computer science comes in first with 62 publications; then social sciences comes in second with 25. Mathematics comes in third with 20 publications, and engineering in fourth with 15 publications. Next is decision sciences with 12 publications, then business, management, and accounting subjects in 5 publications. There are three publications for energy and

materials science, and the last two with two publications are environmental science and medicine.

Publication by research funding

Figure 7 illustrates the representation of research publications based on Research Funding pertaining to utilizing Cloud Computing within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period of time.

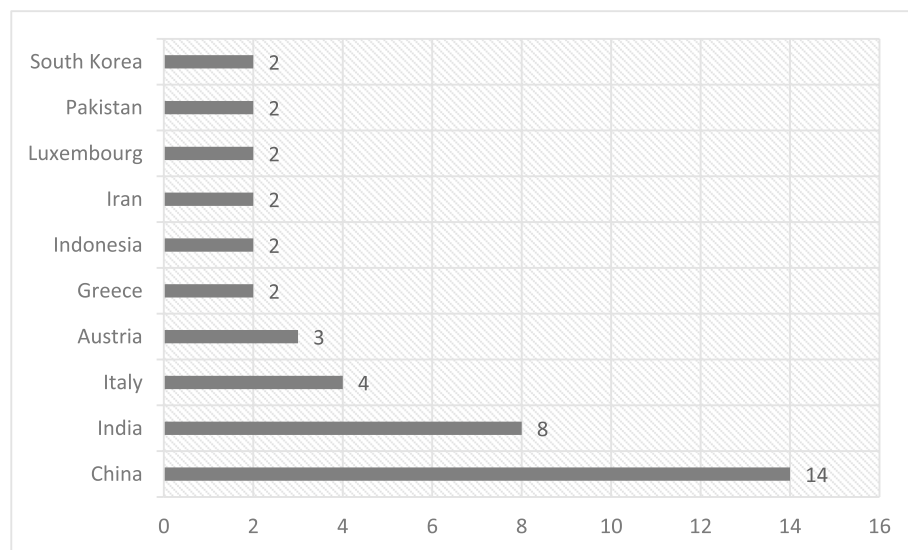


Fig. 5 Publication by countries. Source: scopus database

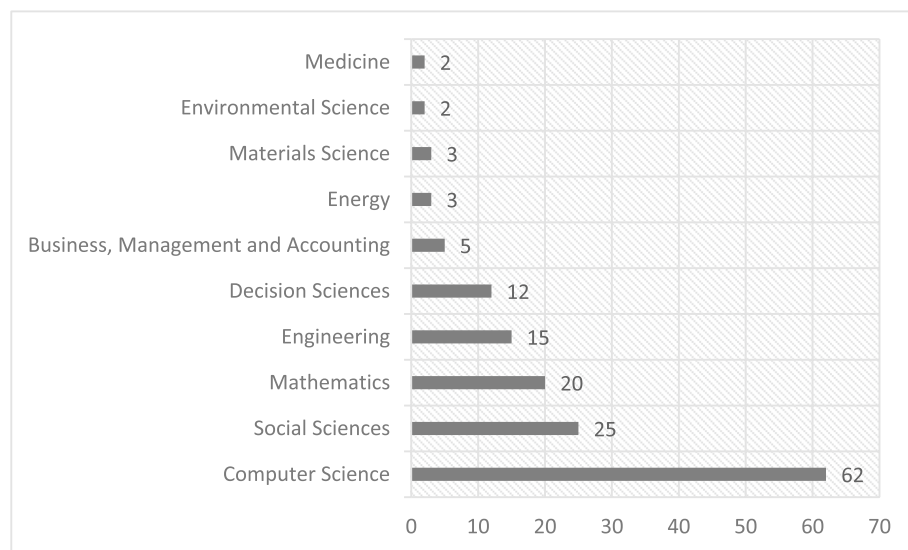


Fig. 6 Publication by venues by subject area. Source: scopus database

Figure 7 shows research funding; there are only three funding: the National Science and Technology Development Agency, which is an agency of the government of Thailand that provides funding for research in the fields of science and technology, as well as the economic applications of such research, National Office for Philosophy and Social Sciences which is a government organization in Beijing, China and Inha University which is a non-public academic institution of higher research that can be found in Incheon, South Korea.

Publication by affiliation

Figure 8 illustrates the representation of research publications based on Affiliation pertaining to the utilization of Cloud Computing within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period of time.

Figure 8 shows ten affiliations with the results of 70 publications with the highest citations. There are three publications: Technische Universitat Graz and Universita

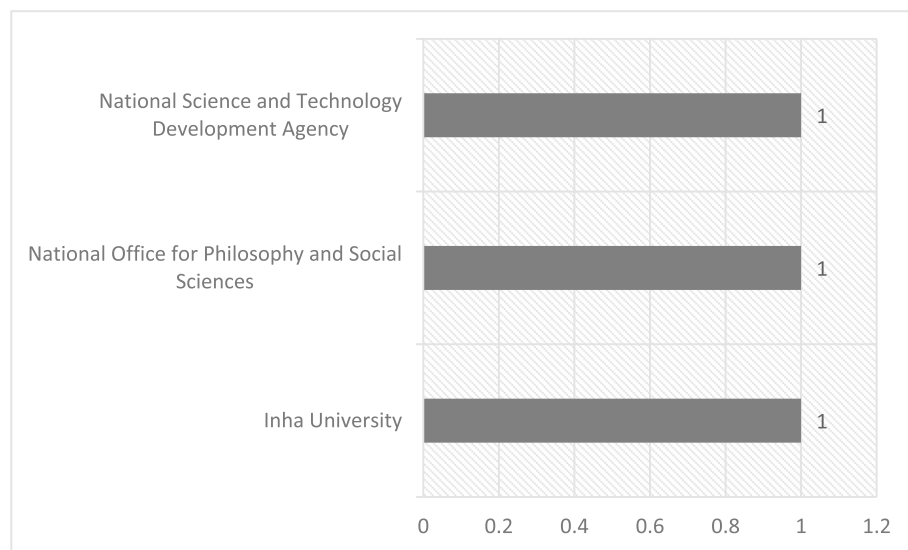


Fig. 7 Publication by research funding. Source: scopus database

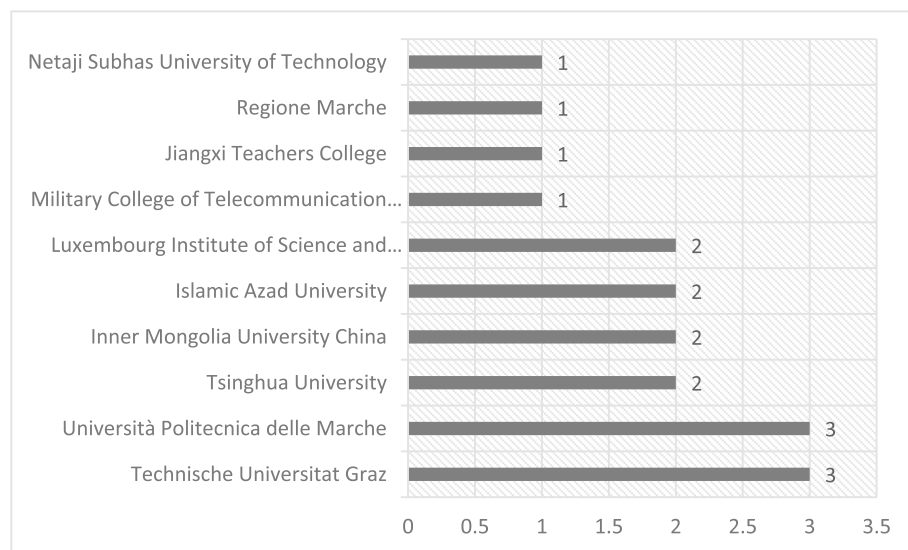


Fig. 8 Publication by affiliation. Source: scopus database

Politecnica delle Marche. Next, there are four affiliations with two publications: Tsinghua University, Inner Mongolia University China, Islamic Azad University, and Luxembourg Institute of Science and Technology. In the last 4, the Military College of Telecommunication Engineering, Jiangxi Teachers College, Region Marche, and Netaji Subhas University of Technology are the affiliations that published one publication.

Publication by source

Figure 9 illustrates the representation of research publications based on Sources pertaining to the utilization of Cloud Computing within the realm of E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period of time.

Figure 9 shows the source, Lecture Notes in Computer Science, Including Subseries Lecture Notes in

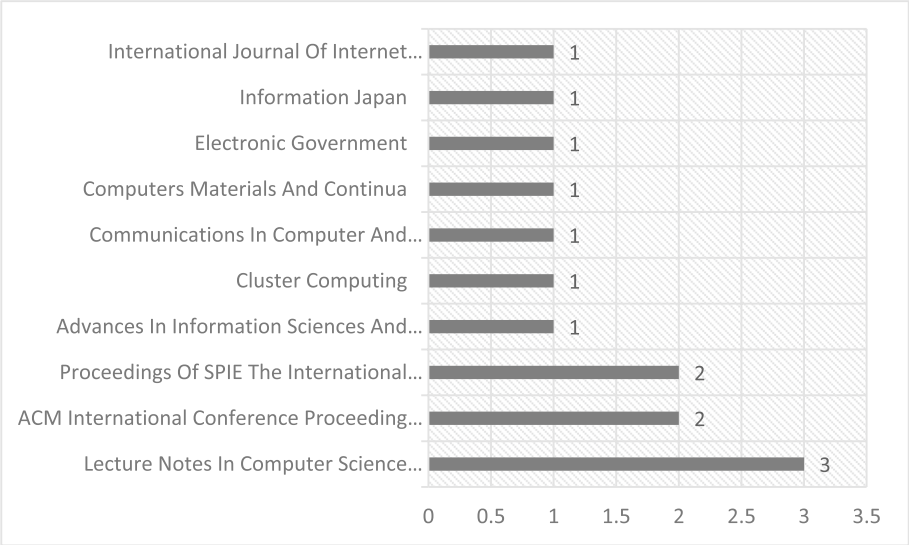


Fig. 9 Publication by source. Source: scopus database

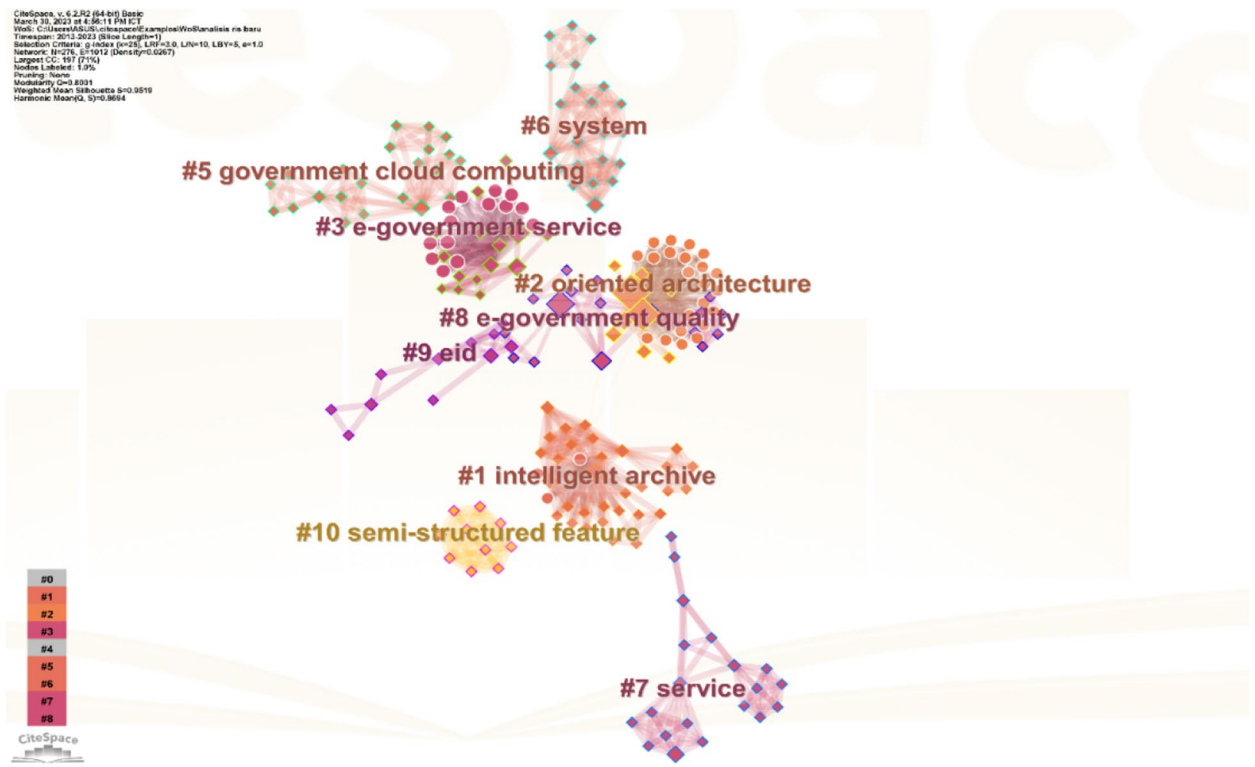


Fig. 10 Cluster analysis. Source: scopus database

Artificial Intelligence and Bioinformatics, which is first cited with three citations. The following 2 with two citations are ACM International Conference Proceeding Series and Proceedings of SPIE, The International

Society for Optical Engineering. With 1 citation are Advances in Information Sciences and Service Sciences, Cluster Computing, Communications in Computer and Information Science, Computers Materials

and Continua, Electronic Government, Information Japan, and International Journal of Internet Technology and Secured Transactions.

Cluster word

Figure 10 illustrates the representation of research publications through Cluster Analysis about using Cloud Computing within E-Government. The subsequent analysis will delve into the patterns and fluctuations observed within the publication trends over a specified period.

Cluster #1 intelligent archive

The cluster word (KTA) shows that 10 clusters represent different keywords. Related to cloud computing, e-government, and architecture, the #1 intelligent archive, relating to those words, comes first. An IA is designed to meet a particular organization's relatively large-scale operational and functional data management and utilization requirements (for example, primary remote sensing data archives, mission- or discipline-specific archives, or specialized commercial archives). Analyses of requirements taken from the context of the precision agricultural use case reveal that the value of a virtual farm comes from its ability to tailor and customize information that is directly relevant to a farm. Intelligent data understanding applications will be required to extract, convert, and show the information a farm requires. The farm will employ this information. Extracting specific information relevant to farming from a mountain of data is crucial to an IA's job. These transformations could involve complicated process chains that involve the reduction of data and the extraction of information or knowledge (Hartberts & Roelofs, 2003).

Cluster #2 oriented architecture

The next cluster would be #2 oriented architecture. The term "service-oriented architecture," or SOA for short, refers to a model for organizing and using dispersed capabilities that may fall under the jurisdiction of various ownership domains [34]. A service-oriented architecture (SOA) aims to create a global mesh of collaborating services published and made available for invocation on a service bus. The adoption of SOA is necessary to fulfill the promises made by Web Services about the agility of businesses and the flexibility of information technology. These benefits are not simply supplied by looking at service architecture from a technological standpoint and adopting Web Service protocols; instead, they necessitate the construction of a Service Oriented Environment that is founded on the following essential characteristics to be effective: The Service Oriented Architecture (SOA) is not only an architecture of services viewed from a technological perspective; instead, it refers to the policies,

processes, and frameworks that we use to ensure that the appropriate services are supplied and consumed. In an SOA environment, it is essential to implement procedures that guarantee the existence of at least two distinct and distinct procedures for the supplier and the client, respectively [35].

Cluster #3 E-government service

The third cluster is the e-government service. E-government can be defined as governments using information and communications technology to improve the overall quality of the services and information offered to their citizens and other stakeholders in a time- and money-efficient manner. The government needs to have a better grasp of how citizens perceive and evaluate the quality of electronically delivered services to continue to provide high-quality services and maintain the confidence of its constituents. According to Parasuraman et al., e-government service quality is affected by system quality and information quality, including ten detailed dimensions: Reliability, Responsiveness, Competence, Access, Courtesy, Communication, Credibility, Safety, Understanding, and Tangibles.

Cluster #5 government cloud computing

The next cluster is cluster #5, government cloud computing. Cloud computing is currently attracting the attention of government agencies worldwide as a potential strategy for improving operational effectiveness while simultaneously cutting expenses. Cloud computing is a model that enables ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (such as networks, servers, storage, applications, and services). These resources can be rapidly provisioned and released with minimal management effort or interaction from service providers. Cloud computing is a model. Cloud computing is a model. Cloud computing has several benefits, including a significant decrease in costs, accessibility from any location, ease of implementation, and provision of the most recent software versions because updates are performed automatically.

Cluster #6 system

Cluster #6 system (in government cloud computing). The cloud gives you access to your information anytime and from any location. The cloud eliminates the need for you to physically be present at the same location as the data storage device. To gain access to the cloud, you will, among other things, need to ensure that you have a working internet connection. This indicates that to see a particular document that you have stored in the cloud, you will first need to connect to the internet using

either a wireless or wired internet connection or a mobile broadband connection, depending on your connection type. The advantage is that you can access the same content regardless of where you are or what device you use as long as it can connect to the internet. These devices could be anything from a desktop computer to a laptop to a tablet to a smartphone.

Cluster #7 service

Cluster #7 service (in government cloud computing). The concept of computing on the cloud is here to stay. It is already in use and can reduce operational expenses, sometimes significantly. They are making it possible to optimize the use of assets and have flexibility in the scale and scope of IT services and hardware². Many government agencies are investigating various new services and interacting with individuals and other entities operating inside the government [36]. Because of its apparent low cost, ease of use, and scalability, cloud computing is poised to transform (if it still needs to do so) how information technology systems are implemented in both the private industry and the public sector. According to the findings of this survey, the primary issues with the implementation of cloud services include ambiguity around the costs of migration, applicable laws and jurisdictions, data safekeeping and privacy, universal connectivity, and applicable laws and jurisdictions over data. It would indicate that most organizations need a comprehensive understanding of the complexities of cloud computing, including insufficient information technology and secure infrastructures for cloud takeoff.

Cluster #8 E-government quality

Cluster #8 e-government quality. The transition to an electronic government is fraught with many challenges. The traditional organizational work culture, the poor quality of human resources in the e-government management, the tight budget limits, and the lack of community recognition are some of the issues that need to be addressed. In the framework of electronic public services, local governments have yet to devote a significant amount of attention to the numerous advantages of the technology. As a direct consequence, the public service model continues to adhere to the traditional modes of operation. The essential step that must be taken is for national and local governments to acknowledge the paradigm change that must take place as they move from establishing e-government in general to e-public service delivery. The government should be open and accountable and work to maximize the benefits and services it provides to the community in the most efficient manner possible. The first thing that must be done is to shift

both the mental model and the bureaucratic work culture within the government.

Cluster #9 Eid

Cluster #9 Eid (in government cloud computing). The term "eIDs," for "electronic identification cards," refers to the digital versions of traditional identification cards used to authenticate a person's identity when conducting business and interacting with others online. On the other hand, a computer in the cloud refers to a technology that facilitates the delivery of computer services through the Internet. Electronic identification and cloud computing have several advantages, including increased safety, accessibility, convenience, and cost-effectiveness. However, installing cloud-based electronic identification systems requires stringent security precautions to protect sensitive personal information. This fact should be noticed. Concerns about privacy and rules designed to protect data must be addressed to guarantee the safe storage and processing of electronic identification information in the cloud.

Cluster #10 semi-structured feature

Cluster #10 semi-structured feature (in government cloud computing). Kiran Kumar Reddi and DnvsI Indira taught us that Big Data combines structured, semi-structured, unstructured, homogeneous, and heterogeneous data. This expanded our understanding of Big Data. The information generally associated with a schema is stored within the data itself in semistructured data, which is why this type of data is often called "self-describing" in some contexts. There is no separate schema in some semistructured data, whereas there is a schema in other forms of semi-structured data. However, it simply limits the data.

Discussion

Examining the pattern of governmental backing for utilizing cloud computing in the architecture of e-government reveals a dynamic scenario influenced by advancing technological progress and administrative priorities. Across the globe, governments are progressively acknowledging the capacity of cloud computing to improve the effectiveness, accessibility, and scalability of e-government services. A noteworthy trend is the increasing tendency toward adopting cloud technology to modernize existing systems and enhance the efficiency of digital governance processes [37]. This transition signifies a broader recognition of cloud technologies' transformative capabilities in optimizing resource use and stimulating innovation within the governmental sector. Furthermore, this trend emphasizes governments' commitment to promoting a digital-first approach to governance, with cloud

computing playing a fundamental role in digital transformation endeavors.

This strategic alignment is motivated by the necessity to meet the escalating expectations of citizens for seamless, user-centered digital interactions while ensuring the safety and dependability of e-government platforms. Consequently, governmental support for utilizing cloud computing encompasses diverse aspects, including policy frameworks, investment incentives, and initiatives for building capacity to facilitate the effective adoption and usage of cloud technology across governmental entities. Moreover, this trend underscores the rise of collaborations between the public and private sectors as a driving force for accelerating the adoption of cloud computing in e-government frameworks [24]. Collaborative endeavors between governments and providers of cloud services support the creation of customized solutions that cater to the public sector's specific needs and regulatory standards. Additionally, such partnerships promote the exchange of knowledge and the enhancement of capabilities, enabling governments to leverage optimal practices and expertise in implementing and managing cloud technology.

Nevertheless, challenges persist, spanning from concerns regarding data security and privacy to issues related to interoperability and the risks of being locked into a particular vendor. Resolving these challenges demands a united effort from policymakers, technologists, and other involved parties to establish strong governance frameworks, standards, and mechanisms to ensure responsibility and transparency in e-government initiatives based on cloud technology [38]. In summary, the trend of governmental support for the utilization of cloud computing in the architecture of e-government signifies a strategic transition towards harnessing digital technologies to provide more agile, citizen-focused public services. Despite the substantial advancements achieved thus far, continuous collaboration and innovation are imperative to fully realize the potential of cloud computing in enhancing efficiency, transparency, and inclusivity in governmental operations.

The discussion of global trends in cloud computing adoption is indeed insightful; however, incorporating a comparative analysis between developed and developing countries would significantly enhance the comprehensiveness of the findings. In developed nations, such as the United States and Germany, cloud computing has been widely embraced, driven by robust technological infrastructure, substantial investment in digital transformation, and a strong regulatory framework that supports innovation. For example, the U.S. government's Cloud First policy encourages federal agencies to adopt cloud solutions, resulting in improved service delivery and

cost savings. In contrast, many developing countries face unique challenges that hinder the widespread adoption of cloud computing. Issues such as limited internet access, inadequate technological infrastructure, and financial constraints often impede progress. For instance, in countries like Nigeria and India, while there is a growing interest in cloud solutions, the lack of reliable connectivity and skilled personnel poses significant barriers to implementation. By juxtaposing these experiences, the analysis can provide a clearer understanding of the factors influencing cloud computing adoption across different contexts, highlighting both the opportunities and challenges that exist globally. This comparative perspective will not only enrich the discussion but also offer valuable insights for policymakers and practitioners seeking to navigate the complexities of cloud computing in e-government.

Implementation strategies

The detailed implementation strategies for integrating cloud computing into e-government systems, it is crucial to outline specific methodologies that governments can adopt. First, a phased implementation approach can be beneficial, where governments begin with pilot projects to test cloud solutions in controlled environments before scaling up. This allows for the identification of potential challenges and the refinement of processes. Second, establishing a comprehensive governance framework is essential, which includes defining roles and responsibilities, setting clear objectives, and ensuring compliance with data protection regulations. Third, investing in training and capacity-building initiatives for government personnel will facilitate a smoother transition to cloud-based systems, ensuring that staff are equipped with the necessary skills to manage and utilize these technologies effectively. Additionally, fostering collaboration with cloud service providers can lead to the development of tailored solutions that meet specific governmental needs while adhering to regulatory standards. By incorporating these detailed strategies, the analysis can provide a more actionable roadmap for governments seeking to implement cloud computing in their e-government initiatives, ultimately enhancing service delivery and citizen engagement.

Challenges and resistance

It is important to provide a more comprehensive discussion on effective strategies for mitigating the potential risks associated with cloud computing in e-government systems. One key challenge is data security; to mitigate this, governments should adopt robust encryption protocols and implement multi-factor authentication to safeguard sensitive information. Additionally, establishing clear data governance policies can help ensure

compliance with privacy regulations and build public trust. Another significant concern is vendor lock-in, which can be addressed by promoting the use of open standards and interoperability frameworks that allow for easier migration between cloud service providers. Furthermore, to tackle the digital divide, governments can invest in digital literacy programs aimed at equipping citizens with the skills necessary to engage with e-government services effectively. Regular risk assessments and audits can also play a crucial role in identifying vulnerabilities and ensuring that cloud systems remain resilient against cyber threats. By elaborating on these mitigation strategies, the analysis can provide a more balanced view that not only acknowledges the challenges but also offers practical solutions for overcoming them, thereby enhancing the overall effectiveness of cloud computing in e-government initiatives.

Successful case studies

It is essential to incorporate empirical evidence and case studies that illustrate the successful implementation of government support for cloud computing in e-government initiatives. For instance, examining case studies from countries like Estonia, which has effectively utilized cloud computing to enhance its digital services, can provide valuable insights into the tangible benefits achieved, such as improved service delivery, increased citizen engagement, and enhanced operational efficiency. Additionally, analyzing data from pilot projects in various jurisdictions can highlight key performance indicators, such as reductions in service response times and cost savings associated with cloud adoption. Furthermore, conducting surveys or interviews with government officials and stakeholders involved in these initiatives can yield qualitative data that reflects the challenges faced and the strategies employed to overcome them. By integrating these empirical findings and real-world examples, the analysis can present a more robust argument for the effectiveness of government support for cloud computing in e-government, thereby demonstrating its potential to drive positive outcomes and inform future policy decisions.

Collaborative mechanisms

In the context of enhancing collaboration among governments, researchers, and practitioners in cloud computing for e-government, it is crucial to identify and define specific platforms and frameworks that can support this cooperation. By delineating these mechanisms, stakeholders can effectively share knowledge, resources, and experiences, which is essential for overcoming challenges and optimizing the implementation of cloud technologies. One effective approach is the establishment of

intergovernmental working groups or task forces. These groups would consist of representatives from various government entities, academic institutions, and industry experts, all focused on the common goal of improving e-government services through cloud computing. By coming together, these stakeholders can share best practices, discuss challenges encountered during implementation, and learn from each other's successes and failures. This collaborative environment fosters innovation and helps to create standardized approaches that can be adopted across different jurisdictions.

To facilitate ongoing communication and collaboration, these working groups can utilize online collaboration platforms. Such platforms can include dedicated forums, shared document repositories, and project management tools that allow members to exchange ideas, share research findings, and develop joint initiatives. By leveraging technology, these platforms can enable real-time discussions and provide a space for continuous engagement, regardless of geographical barriers. This structured approach to collaboration not only enhances the collective knowledge base but also accelerates the development and implementation of effective cloud solutions in e-government, ultimately leading to improved public services and citizen engagement.

Policy development process

The formulation and implementation of policies related to cloud computing in e-government are critical processes that require careful planning and execution. Initially, the policy development process begins with a thorough assessment of the current landscape. This involves analyzing existing regulations that govern data management, privacy, and safety, as well as evaluating the technological capabilities of government agencies. Understanding stakeholder needs is also paramount, as it ensures that the policies developed are relevant and address the actual challenges faced by users, including government officials, industry players, and citizens. Following this assessment, stakeholder consultations play a vital role in the process. Engaging with a diverse group of participants allows for the collection of valuable insights and perspectives, which helps to identify key priorities and challenges that need to be addressed in the policy proposals. This collaborative approach fosters a sense of ownership among stakeholders and increases the likelihood of successful policy adoption.

Once a foundational understanding is established, a drafting committee is formed to create policy proposals that align with the identified needs and objectives. These proposals are then subjected to a public review process, where feedback from a broader audience is solicited. This step is crucial for refining the policies, as it allows

for the incorporation of diverse viewpoints and ensures that the final policies are well-rounded and effective. The implementation phase of these policies is multifaceted. It includes establishing clear guidelines that outline the roles and responsibilities of various stakeholders, developing training programs to equip government personnel with the necessary skills, and allocating resources to facilitate the transition to cloud-based systems. Furthermore, integrating ongoing monitoring and evaluation mechanisms is essential to assess the effectiveness of the policies over time. This allows for timely adjustments to be made in response to emerging technologies and evolving public needs.

By elaborating on these steps, the analysis provides a comprehensive view of the policy development process. It underscores the importance of collaboration, transparency, and adaptability in creating effective frameworks for cloud computing in e-government. This thorough understanding not only addresses the gaps in the existing literature but also enhances the credibility and applicability of the proposed policies, ultimately contributing to more effective governance and improved public services.

Conclusion

Based on the analysis of bibliometric government support for cloud computing e-government architecture in 2023, several key conclusions can be drawn: **Increasing Emphasis on Cloud Computing:** The analysis indicates a growing trend of government support for cloud computing in the context of e-government architecture. Governments recognize the potential of cloud computing in enhancing the efficiency, scalability, and accessibility of e-government services; **Adoption of E-Government Architecture:** Governments are actively adopting e-government architecture to improve service delivery, streamline administrative processes, and foster citizen engagement. The analysis suggests that cloud computing is crucial in supporting and enabling e-government architecture; **Strong Government Backing:** The bibliometric analysis reveals significant research and publications on government support for cloud computing in e-government architecture. This indicates a firm commitment from governments to explore and implement cloud-based solutions in the context of e-governance.

Collaboration and Knowledge Sharing: The analysis highlights a collaborative approach among governments, researchers, and practitioners in sharing knowledge, best practices, and lessons learned regarding cloud computing in e-government architecture. This collaborative effort is essential for fostering innovation, avoiding duplication of efforts, and promoting effective implementation strategies; **Policy Development and Standardization:** The analysis suggests that governments are actively involved

in developing policies and standards related to cloud computing in e-government architecture. This indicates a recognition of the need for regulatory frameworks to ensure data security, privacy, interoperability, and vendor neutrality in cloud-based e-government systems.

It is essential to provide a more nuanced analysis of how various governments are approaching the adoption of cloud computing in their e-government architecture. Different countries exhibit distinct strategies influenced by their unique socio-economic contexts, regulatory environments, and technological infrastructures. For instance, some governments prioritize public-private partnerships to leverage expertise and resources from cloud service providers, while others focus on developing in-house capabilities to maintain control over sensitive data. Additionally, the pace of cloud adoption varies significantly; advanced economies may rapidly implement sophisticated cloud solutions, whereas developing nations might adopt a more gradual approach, addressing foundational issues such as digital literacy and infrastructure gaps first. By examining these diverse strategies, we can gain a deeper understanding of the factors that shape cloud computing adoption in e-government, highlighting best practices and potential pitfalls that different governments encounter along the way. This detailed exploration will enrich the conclusions and provide actionable insights for policymakers and stakeholders involved in e-government initiatives.

Potential Challenges and Risks: While the analysis emphasizes the positive aspects of government support for cloud computing in e-government architecture, it is essential to acknowledge potential challenges and risks. These may include concerns related to data privacy, data breaches, vendor lock-in, and the digital divide among citizens. Overall, the analysis of bibliometric government support for cloud computing e-government architecture in 2023 indicates a positive trajectory, with governments actively embracing cloud technologies to enhance their e-governance initiatives. However, policymakers and stakeholders must address potential challenges and risks to ensure the successful and sustainable implementation of cloud-based e-government systems.

Limitations and implications

While the advantages of cloud computing in e-government are well-documented, it is equally important to address the counterarguments and limitations associated with its adoption. One significant concern is the dependence on cloud service providers, which can lead to potential vendor lock-in, where governments may find it challenging to switch providers due to proprietary technologies or contractual obligations. This situation can limit flexibility and increase costs over time. Additionally,

the feasibility of adopting cloud solutions in resource-constrained environments poses a challenge, as not all local governments may have the necessary infrastructure or technical expertise to implement these systems effectively. For instance, a study by Al Mudawi et al. [37] highlights successful implementations of cloud-based e-government services in countries like Estonia, where a robust digital infrastructure has enabled seamless service delivery. However, in contrast, many developing nations struggle with inadequate internet connectivity and limited financial resources, which can hinder the effective deployment of cloud solutions. By acknowledging these limitations and providing examples of both successful and challenging implementations, the paper can present a more balanced view of cloud computing's role in e-government, ultimately enhancing the credibility of its claims and recommendations.

While this study provides valuable insights into the trends of cloud computing in e-government through bibliometric analysis, it is essential to address certain challenges and limitations inherent in this approach. One potential bias in bibliometric analysis is the reliance on published literature, which may not fully capture the breadth of ongoing research or practical implementations in the field. This can lead to an overrepresentation of certain perspectives while neglecting emerging voices or innovative practices that are not yet documented in academic publications. Additionally, the use of CiteSpace, while a powerful tool for visualizing citation networks and identifying research hotspots, has its limitations. For instance, CiteSpace's algorithms may prioritize more frequently cited works, potentially overshadowing less popular but equally significant studies that contribute to the discourse on cloud computing in e-government. Furthermore, the software's reliance on specific databases may introduce biases based on the availability and accessibility of literature within those platforms. Acknowledging these limitations is crucial for contextualizing the findings and ensuring that future research endeavors consider a more holistic view of the landscape, incorporating qualitative insights and diverse sources of information to complement the quantitative data derived from bibliometric analysis.

Future recommendation

It can be further strengthened by incorporating actionable recommendations for policymakers and practitioners. To facilitate the successful integration of cloud computing, policymakers should prioritize the development of comprehensive frameworks that address data security, privacy, and interoperability standards. This

includes establishing clear guidelines for cloud service procurement and fostering partnerships with reputable cloud providers to mitigate risks associated with vendor lock-in. Additionally, training programs should be implemented to build the digital skills of government personnel, ensuring they are equipped to manage and utilize cloud technologies effectively. Practitioners should also be encouraged to adopt a phased approach to cloud implementation, starting with pilot projects that allow for testing and refinement before full-scale deployment. Furthermore, fostering collaboration between government agencies, private sector partners, and academic institutions can promote knowledge sharing and innovation in cloud-based solutions. By taking these actionable steps, stakeholders can enhance the effectiveness and sustainability of cloud computing initiatives in e-government, ultimately leading to improved public service delivery and citizen engagement.

As technology continues to evolve, future research could explore the implications of artificial intelligence and machine learning integration with cloud services, particularly in enhancing data analytics and decision-making processes within government operations. Additionally, investigating the impact of emerging technologies, such as blockchain, on cloud computing security and transparency in e-government could provide valuable insights into safeguarding citizen data. Another critical area for future research is the examination of the digital divide, particularly how disparities in access to cloud technologies affect service delivery in underserved communities. Understanding these dynamics can inform strategies to promote inclusivity and equitable access to e-government services. Furthermore, longitudinal studies that track the long-term effects of cloud adoption on government efficiency and citizen satisfaction would contribute to a deeper understanding of the benefits and challenges associated with cloud computing in the public sector. By addressing these emerging trends and unresolved issues, future research can play a pivotal role in shaping the ongoing discourse and guiding effective policy and practice in the realm of cloud-based e-government.

Authors' contributions

M.Y. and E.P.P. wrote the main manuscript text. F.M. and T.K. prepared figures. A.N., D.M., and H.A.M. fixed reviewer comments. All authors reviewed the manuscript.

Funding

No Funding is provided for this research.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Received: 11 May 2024 Accepted: 6 February 2025

Published online: 25 February 2025

References

- Niswar M, Lawi A, Teknik F, Timur UI, Teknik F, Hasanuddin U (2014) Kakas kolaborasi e-government berbasis cloud computing. 3(1):87–91
- Raman A (2016) How do social media, mobility, analytics and cloud computing impact nonprofit organizations? A pluralistic study of information and communication technologies in indian context. *Inf Technol Dev* 22(3):400–421. <https://doi.org/10.1080/02681102.2014.992002>
- Zwattendorfer B, Stranacher K, Tauber A, Reichstädter P (2013) Cloud computing in e-government across Europe a comparison. *Lect Notes Comput Sci*. (including Subser Lect Notes Artif Intell Lect Notes Bioinformatics). 8061 LNCS:181–195. https://doi.org/10.1007/978-3-642-40160-2_15
- Mulyono T, Tiara Devani F, Ayu Puspitaningrum E, Qorih Putri V, Sabtiana R (2018) E-Government Berbasis Cloud Computing Pada Pemerintah Daerah. *Jurnal Tike* 1
- Davidavičienė V, Raudeliūnienė J, Vengrienė E, Jakubavičius A (2018) Consolidation of the Activities of Regulatory Institutions While Implementing E-Government Solutions. *J Bus Econ Manage* 19(2):307–22. <https://doi.org/10.3846/jbem.2018.5534>
- Setiawan W, Fajriyah N, Duha T (2022) Analisa Layanan cloud computing di era digital. *J Inform* 1(1):32–39
- Sidiq M (2019) Panduan Analisis Bibliometrik Sederhana. no. June. <https://doi.org/10.13140/RG.2.2.15688.37125>
- Mohammed F, Alzahrani AI, Alfarraj O, Ibrahim O (2017) Cloud computing fitness for E-government implementation: importance-performance analysis. *IEEE Access* 6:1236–1248. <https://doi.org/10.1109/ACCESS.2017.2778093>
- Ahmed W, Wu YW (2013) A Survey on Reliability in Distributed Systems. *J Comput SystSci* 79(8):1243–1255. <https://doi.org/10.1016/j.jcss.2013.02.006>
- Lesjak B, Pšenica I (2022) Perception of ICT Use by Residents of Smart Cities. *Int J Innov Learn* 31(4):459–473. <https://doi.org/10.1504/IJIL.2022.123174>
- Faiq F (2018) Implementasi Cloud Computing Di Beberapa Instansi Pemerintahan. *Masyarakat Telematika Dan Informasi : Jurnal Penelitian Teknologi Informasi Dan Komunikasi* 8(2):97. <https://doi.org/10.17933/mti.v8i2.105>
- Fauziah F (2014) Arsitektur Cloud Computing Pada Sistem Informasi Desa Sebagai Layanan Akses Informasi Desa. *Semin. Nas. Inform.* 2014(semnasIF):224–230
- Irma I, Ferdiana R, Nugroho LE (2017) Implementasi Cloud Computing Sebagai Digital Asset Management (DAM) Di Istana Kepresidenan Yogyakarta. *Prosiding Seminar Nasional Geotik*, p 2580–8796. <https://publikasi.lmiah.ums.ac.id/xmlui/handle/11617/9094>
- Dhaoui I (2021) E-Government for Sustainable Development: Evidence from MENA Countries. *J Knowl Econ* 13(3):2070–2099. <https://doi.org/10.1007/s13132-021-00791-0>
- Al Mudawi N, Beloff N, White M (2020) Issues and Challenges: Cloud Computing e-Government in Developing Countries. *Int J Adv Comput Sci Appl* 11(4):7–11. <https://doi.org/10.14569/IJACSA.2020.0110402>
- Al Mudawi N, Beloff N, White M (2020) Issues and challenges: Cloud computing e-government in developing countries. *Int J Adv Comput Sci Appl*. 11(4):7–11. <https://doi.org/10.14569/IJACSA.2020.0110402>
- Hashemi S (2013) Cloud computing technology for Egovernment architecture. *Int J Found Comput Sci Technol* 3(6):15–23. <https://doi.org/10.5121/ijfst.2013.3602>
- Ji H, Liang Y (2016) Exploring the Determinants Affecting E-Government Cloud Adoption in China. *Int J Bus Manage* 11(4):81. <https://doi.org/10.5539/ijbm.v11n4p81>
- Ma S, Hao F, Lin Y, Liang Y (2022) The Construction of Big Data Computational Intelligence System for E-Government in Cloud Computing Environment and Its Development Impact. *Comput Intell Neurosci*. 2022. <https://doi.org/10.1155/2022/7295060>
- Wang H, Singhal A, Liu P (2023) Tackling Imbalanced Data in Cybersecurity with Transfer Learning: A Case with ROP Payload Detection. *Cybersecurity* 6(1):2. <https://doi.org/10.1186/s42400-022-00135-8>
- Alkhwaldi A, Kamala M, Qahwaji R (2018) Analysis of Cloud-Based E-Government Services Acceptance in Jordan: Challenges and Barriers. *J Internet Technol Secured Trans* 6(2):556–568. <https://doi.org/10.20533/jitst.2046.3723.2018.0069>
- Karim F (2022) Cloud computing-based m-government. *Inform.* 46(5):69–73. <https://doi.org/10.31449/inf.v46i5.3879>
- ALMutairi NN, Fahad Thuwaini S (2015) Cloud Computing Uses for E-Government in the Middle East Region Opportunities and Challenges. *Int J Bus Manag.* 10(4). <https://doi.org/10.5539/ijbm.v10n4p60>
- Siam AI et al (2021) Secure health monitoring communication systems based on IoT and cloud computing for medical emergency applications. *Comput Intell Neurosci* 2021:8016525. <https://doi.org/10.1155/2021/8016525>
- Hwang K, Um H (2021) Social Controls and Bonds of Public Information Consumer on Sustainable Utilization and Provision for Computing. *Sustainability (Switzerland)* 13(9). <https://doi.org/10.3390/su13095263>
- Patel SK (2023) Improving Intrusion Detection in Cloud-Based Healthcare Using Neural Network. *Biomed Signal Process Control* 83. <https://doi.org/10.1016/j.bspc.2023.104680>
- Zhang Y, Kimathi FA (2022) Exploring the Stages of E-Government Development from Public Value Perspective. *Technol Soc* 69. <https://doi.org/10.1016/j.techsoc.2022.101942>
- Almarabeh T, Majdalawi YK, Mohammad H (2016) Cloud computing of E-government. *Commun Netw* 08(01):1–8. <https://doi.org/10.4236/cn.2016.81001>
- Looi CK, Wong LH, Biswas G, Du Boulay B, Hoppe HU, Mizoguchi R (2018) ‘Smartness of Technology’ or ‘Smartness of Learners’: Is It a Zero-Sum Game? Or Can We Have Both? In: Rodrigo MMT, Yang J-C, Wong L-H, Chang M (eds) ICCE 2018 - 26th International Conference on Computers in Education, Main Conference Proceedings. Learning Sciences Lab., National Institute of Education, Nanyang Technological University, Singapore. Asia-Pacific Society for Computers in Education, Singapore, p i36–i42. <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85060004350&partnerId=40&md5=5f3fd38c25e3d1409ea9f8f30bb19c70>
- Maharana B, Mahapatra S (2006) Google Scholar : A Tool To Search Scholarly Information on the Web. *Search*, 09–10
- Burnham JF (2006) Scopus Database: A Review. *Biomed Digit Libr* 3:1–8. <https://doi.org/10.1186/1742-5581-3-1>
- Liu Z, Ren L, Li R, Liu Q, Zhao Y (2022) ID-Based Sanitizable Signature Data Integrity Auditing Scheme with Privacy-Preserving. *Comput Secur* 121. <https://doi.org/10.1016/j.cose.2022.102858>
- Taşkın Z, Aydinoglu AU (2015) Collaborative Interdisciplinary Astrobiology Research: A Bibliometric Study of the Nasa Astrobiology Institute. *Scintometrics* 103(3):1003–22. <https://doi.org/10.1007/s11192-015-1576-8>
- Söderström E, Henkel M, Perjons E (2011) Reference Models for Service Oriented Architectures. *Proceedings of the IADIS International Conference Information Systems 2011*, IS 2011, no. October: 187–95
- Matters, Process, and The Service Architecture (2009) *Understanding Service-Oriented Architecture*
- Lee S, Choi Y, Ra J, Kim J, Ashihara K (2020) Impact of public cloud computing service in korean government organizations. *ICIC Express Lett Part B Appl*. 11(3):313–318. <https://doi.org/10.24507/icicelb.11.03.313>
- Al Mudawi N, Beloff N, White M (2019) Cloud computing in government organizations-towards a new comprehensive model. In 2019 IEEE SmartWorld, Ubiquitous Intelligence and Computing, Advanced and Trusted Computing, Scalable Computing and Communications, Internet of People and Smart City Innovation, SmartWorld/UIC/ATC/SCALCOM/IOP/SCI 2019. 1473–1479. <https://doi.org/10.1109/SmartWorld-UIC-ATC-SCALCOM-IOP-SCI.2019.00266>
- Mohammed F, Ibrahim (2013) Refining E-government Readiness Index by Cloud Computing. *J Teknol (Sciences Eng)*. 65(1):23–34. <https://doi.org/10.11113/jt.v65.1759>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.